Closed-Form Observed Information for Coordinate-Based Meta-Regression

Derivations, equivalence proofs, and the structure of the covariance step

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Abstract

NiMARE's coordinate-based meta-regression (CBMR) computes regression standard errors by differentiating the log-likelihood twice with `torch.func.hessian`. Forward-over-reverse mode pushes every tangent through the likelihood at once, so the intermediate has shape (parameters × patterns × voxels); at 7,329 coefficients, 21 patterns and 8,444 voxels that is 9.7 GB, which is an out-of-memory failure rather than a slow fit.

We derive the observed information in closed form for all three of NiMARE's observation distributions, prove each derivation equivalent to the autodiff Hessian, and characterise the sparsity the design induces in it. A single lemma covers all three: the second differential of the log-likelihood with respect to the log-intensity is either diagonal in the voxels (Poisson, Negative Binomial) or diagonal-plus-rank-one (Clustered Negative Binomial), and the chain rule through the log-bilinear predictor turns each into n_{patterns} symmetric rank-updates of width n_{bases}. No parameter-sized intermediate is ever formed. We then separate two questions that are easily conflated: which *likelihood* admits a closed-form Hessian (all three), and which *designs* make that Hessian block diagonal (a strictly narrower class), and we give the generalisations that recover the fast covariance step outside it.

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1 Setting and notation

1.1 The pattern representation

CBMR models the foci of n experiments over V voxels as a point process whose log-intensity is bilinear in a spatial spline basis and an experiment-level design. Write

- $B \in \mathbb{R}^{V\times K}$ for the spline basis, $b_v^\top$ its v-th row;
- $L \in \mathbb{R}^{P\times C}$ for the *loadings*: the distinct rows of the experiment-level blocks of every spatial term, stacked. P is the number of distinct spatial loadings ("patterns"), C the number of spatial columns;
- $\pi : \{1,\dots,n\} \to \{1,\dots,P\}$ for the pattern of each experiment, and $\mathcal{I}_p = \pi^{-1}(p)$;
- $\beta \in \mathbb{R}^{C\times K}$ for the spatial coefficients;
- $G \in \mathbb{R}^{n\times Q}$ for the non-spatial ("global") design block and $\gamma \in \mathbb{R}^Q$ for its coefficients.

The log-intensity of pattern p at voxel v, and the scalar moderator effect of experiment i, are

$$S_{pv} \;=\; \sum_{c=1}^{C} L_{pc} \sum_{k=1}^{K} \beta_{ck} B_{vk}, \qquad m_i \;=\; \sum_{q=1}^{Q} G_{iq}\gamma_q, \tag{1}$$

so that $S = (L\beta)B^\top$ and $m = G\gamma$. We write $u_{pv} = e^{S_{pv}}$ and $e_i = e^{m_i}$ throughout. Foci counts are y_{iv}, with pattern marginals and experiment totals

$$Y_{pv} \;=\; \sum_{i\in\mathcal{I}_p} y_{iv}, \qquad y_{i\cdot} \;=\; \sum_{v=1}^{V} y_{iv}. \tag{2}$$

The parameter vector is $\phi = (\mathrm{vec}(\beta), \gamma) \in \mathbb{R}^{CK+Q}$, with β flattened row-major so that β_{ck} occupies index $cK + k$. Overdispersion parameters θ_p, where the distribution has them, are held fixed at their fitted values — this is what `CBMRModel.information_matrix` does, and it is what CBMR has always reported regression standard errors from. Every derivation below is therefore a differentiation in ϕ only, with θ a constant.

1.2 The two facts the whole paper rests on

Both parts of (1) are *linear* in the parameters. Hence

$$\frac{\partial S_{pv}}{\partial \beta_{ck}} = L_{pc} B_{vk}, \qquad \frac{\partial^2 S_{pv}}{\partial \beta_{ck}\, \partial \beta_{c'k'}} = 0, \qquad \frac{\partial m_i}{\partial \gamma_q} = G_{iq}, \qquad \frac{\partial^2 m_i}{\partial \gamma_q\, \partial \gamma_{q'}} = 0. \tag{3}$$

The second derivatives vanish identically. Everything that follows is the chain rule applied to this fact, so the only distribution-specific work is the second differential of the log-likelihood with respect to (S, γ).

2 A master lemma

Every CBMR log-likelihood in NiMARE decomposes over patterns,

$$-\log \mathcal{L}(\phi) \;=\; \sum_{p=1}^{P} f_p\big(S_{p\cdot},\, \gamma\big), \tag{4}$$

because experiments sharing a pattern share a log-intensity map, and no term couples two patterns. This is the representation `predictor.py` is built on.

Lemma 1 (Chain rule through a log-bilinear predictor)**.** *Let $F(\phi) = \sum_p f_p(S_{p\cdot}, \gamma)$ with S as in* (1) *and each f_p twice differentiable. Write*

$$\big(\Sigma_p\big)_{vv'} = \frac{\partial^2 f_p}{\partial S_{pv} \partial S_{pv'}}, \qquad \big(\tau_p\big)_{vq} = \frac{\partial^2 f_p}{\partial S_{pv} \partial \gamma_q}, \qquad \big(\Xi_p\big)_{qq'} = \frac{\partial^2 f_p}{\partial \gamma_q \partial \gamma_{q'}}.$$

Then the Hessian of F has blocks

$$\frac{\partial^2 F}{\partial \beta_{ck}\, \partial \beta_{c'k'}} = \sum_{p=1}^{P} L_{pc} L_{pc'} \left(B^\top \Sigma_p B\right)_{kk'}, \tag{5}$$

$$\frac{\partial^2 F}{\partial \beta_{ck}\, \partial \gamma_q} = \sum_{p=1}^{P} L_{pc} \left(B^\top \tau_p\right)_{kq}, \tag{6}$$

$$\frac{\partial^2 F}{\partial \gamma_q\, \partial \gamma_{q'}} = \sum_{p=1}^{P} \big(\Xi_p\big)_{qq'}. \tag{7}$$

Proof. By (3), $\partial S_{pv}/\partial \beta_{ck} = L_{pc} B_{vk}$ and the second derivative of S vanishes, so the chain rule for a composition with an affine map contributes no curvature term:

$$\frac{\partial^2 f_p}{\partial \beta_{ck} \partial \beta_{c'k'}} = \sum_{v,v'} \frac{\partial^2 f_p}{\partial S_{pv} \partial S_{pv'}} \frac{\partial S_{pv}}{\partial \beta_{ck}} \frac{\partial S_{pv'}}{\partial \beta_{c'k'}} = L_{pc} L_{pc'} \sum_{v,v'} (\Sigma_p)_{vv'} B_{vk} B_{v'k'},$$

which is (5). The same argument with one factor replaced by $\partial \gamma_q / \partial \gamma_q = 1$ gives (6), and with both replaced, (7). Summing over p completes the proof. □

Lemma 1 reduces each distribution to a single question: *what is Σ_p?* The answer determines the cost. If $\Sigma_p = \operatorname{diag}(w_p)$ then $B^\top \Sigma_p B = B^\top \operatorname{diag}(w_p) B$ costs $O(VK^2)$ and allocates one $V \times K$ temporary. If Σ_p is diagonal plus rank one, one extra outer product of a K-vector is added. In neither case is anything of size CK formed.

Remark 2. This is exactly what forward-over-reverse autodiff fails to exploit. `jacfwd` carries one tangent per parameter through the evaluation, materialising an array of shape (CK, P, V), because it cannot know that the map $\beta \mapsto S$ is affine with a rank structure that collapses the parameter axis.

3 Poisson

3.1 The likelihood

Dropping the parameter-free $-\sum \log y!$, NiMARE's Poisson log-likelihood on marginals is

$$\log \mathcal{L} \;=\; \sum_p \sum_v Y_{pv} S_{pv} \;+\; \sum_i y_{i\cdot}\, m_i \;-\; \sum_p \Big(\sum_v u_{pv}\Big) T_p, \qquad T_p = \sum_{i \in \mathcal{I}_p} e_i. \tag{8}$$

Proposition 3 (Poisson observed information)**.** *With $U_{pq} = \sum_{i \in \mathcal{I}_p} e_i G_{iq}$, $a_{pk} = \sum_v u_{pv} B_{vk}$ and $E_p = \sum_v u_{pv}$, the Hessian of $-\log \mathcal{L}$ is*

$$H_{(ck),(c'k')} = \sum_p L_{pc} L_{pc'} \big(B^\top \operatorname{diag}(T_p u_{p\cdot}) B\big)_{kk'}, \tag{9}$$

$$H_{(ck),q} = \sum_p L_{pc}\, a_{pk} U_{pq}, \tag{10}$$

$$H_{q,q'} = \sum_i e_i E_{\pi(i)} G_{iq} G_{iq'}. \tag{11}$$

In particular H does not depend on the foci.

Proof. Write (8) in the form (4) with $f_p = -\sum_v Y_{pv} S_{pv} - \sum_{i \in \mathcal{I}_p} y_{i\cdot} m_i + E_p(S_{p\cdot})\, T_p(\gamma)$. The first two terms are linear in ϕ and so contribute nothing to any second derivative; this is the whole of the data dependence, which proves the final claim. For the remainder, $\partial E_p / \partial S_{pv} = u_{pv}$ and $\partial^2 E_p / \partial S_{pv} \partial S_{pv'} = u_{pv} \delta_{vv'}$, while $\partial T_p / \partial \gamma_q = U_{pq}$ and $\partial^2 T_p / \partial \gamma_q \partial \gamma_{q'} = \sum_{i \in \mathcal{I}_p} e_i G_{iq} G_{iq'}$. Hence $\Sigma_p = T_p \operatorname{diag}(u_{p\cdot})$, $(\tau_p)_{vq} = u_{pv} U_{pq}$ and $(\Xi_p)_{qq'} = E_p \sum_{i \in \mathcal{I}_p} e_i G_{iq} G_{iq'}$. Substituting into Lemma 1 and collecting the $\gamma\gamma$ sum over experiments gives the three displays. □

Remark 4. The foci-independence is the log link's canonical-parameter property: for a GLM with canonical link the observed and expected information coincide. It is a useful test of an implementation, since a Poisson Hessian that responds to the data is wrong; the test suite checks this.

4 Negative Binomial

4.1 The likelihood in (η_p, κ_p)

NiMARE's negative binomial is a moment-matched approximation to the sum, over the experiments of a pattern, of negative binomial variables. With overdispersion θ_p and

$$s_1 = \sum_{i \in \mathcal{I}_p} e_i, \qquad s_2 = \sum_{i \in \mathcal{I}_p} e_i^2, \tag{12}$$

it uses shape $r = s_1^2 / (\theta_p s_2)$ and success probability $\pi_v = \big(1 + s_1/(\theta_p u_{pv} s_2)\big)^{-1}$. That is, Y_{pv} is negative binomial with mean $\mu_{pv} = s_1 u_{pv}$ and dispersion $k_p = \theta_p s_2 / s_1^2$, so that $\operatorname{Var} Y_{pv} = \mu_{pv} + k_p \mu_{pv}^2$ and $r = 1/k_p$. We work in

$$\eta_p \;=\; \log s_1, \qquad \kappa_p \;=\; \log k_p, \qquad x_{pv} \;=\; k_p \mu_{pv}, \tag{13}$$

which give $\pi_v = x_{pv}/(1 + x_{pv})$ and $1 - \pi_v = 1/(1 + x_{pv})$.

Lemma 5. *Up to an additive constant depending only on the data, $f_p = -\sum_v \lambda(\eta_p, \kappa_p, S_{pv}; Y_{pv})$ with*

$$\lambda(\eta, \kappa, S; Y) \;=\; \log\Gamma(Y + e^{-\kappa}) - \log\Gamma(e^{-\kappa}) + Y\kappa + Y(\eta + S) - (Y + e^{-\kappa}) \log\big(1 + e^{\kappa + \eta + S}\big). \tag{14}$$

Proof. Substituting $r = e^{-\kappa}$, $\log(1-\pi_v) = -\log(1+x_{pv})$ and $\log \pi_v = \kappa_p + \eta_p + S_{pv} - \log(1+x_{pv})$ into NiMARE's summand $\log\Gamma(Y+r) - \log\Gamma(Y+1) - \log\Gamma(r) + r\log(1-\pi) + Y\log\pi$ gives (14) minus $\log\Gamma(Y_{pv}+1)$, which is free of ϕ. □

The value of (14) is that it separates the three ways the parameters enter: the special functions see only κ_p, the voxels enter only through $\eta_p + S_{pv}$, and the moderators only through (η_p, κ_p). Every derivative in S_{pv} is therefore the corresponding derivative in η_p.

Proposition 6 (Negative binomial observed information)**.** *Write*

$$w_{pv} \;=\; \frac{\mu_{pv} + Y_{pv}x_{pv}}{(1+x_{pv})^2}, \qquad c_{pv} \;=\; \frac{(Y_{pv} - \mu_{pv})\,x_{pv}}{(1+x_{pv})^2}, \qquad g(x) \;=\; \frac{\log(1+x)}{x} - \frac{1}{1+x}, \tag{15}$$

and, for integer counts, with n_{pj} the number of voxels at which $Y_{pv} > j$,

$$G_{1p} = \sum_{j\ge 0} n_{pj}\frac{jk_p}{1+jk_p}, \qquad G_{2p} = \sum_{j\ge 0} n_{pj}\frac{jk_p}{(1+jk_p)^2}. \tag{16}$$

Then $\Sigma_p = \mathrm{diag}(w_{p\cdot})$, $(\tau_p)_{vq} = w_{pv}\,\partial_q\eta_p + c_{pv}\,\partial_q\kappa_p$, and

$$\begin{aligned}(\Xi_p)_{qq'} \;=\; &\Big(\sum_v w_{pv}\Big)\partial_q\eta_p\,\partial_{q'}\eta_p + \Big(\sum_v c_{pv}\Big)\big(\partial_q\eta_p\,\partial_{q'}\kappa_p + \partial_{q'}\eta_p\,\partial_q\kappa_p\big) - \Lambda_{\kappa\kappa}\,\partial_q\kappa_p\,\partial_{q'}\kappa_p \\ &- \Lambda_\eta\,\partial^2_{qq'}\eta_p - \Lambda_\kappa\,\partial^2_{qq'}\kappa_p,\end{aligned} \tag{17}$$

where

$$\begin{aligned}\Lambda_\eta &= \sum_v \frac{Y_{pv} - \mu_{pv}}{1+x_{pv}}, \\ \Lambda_\kappa &= G_{1p} + \sum_v \Big(\mu_{pv}\,g(x_{pv}) - \frac{Y_{pv}x_{pv}}{1+x_{pv}}\Big), \\ \Lambda_{\kappa\kappa} &= G_{2p} + \sum_v \Big(\mu_{pv}\Big(\frac{x_{pv}}{(1+x_{pv})^2} - g(x_{pv})\Big) - \frac{Y_{pv}x_{pv}}{(1+x_{pv})^2}\Big).\end{aligned} \tag{18}$$

Proof. Differentiate (14), using $\partial\mu/\partial\eta = \mu$, $\partial x/\partial\eta = \partial x/\partial\kappa = x$ and $e^{-\kappa}x = \mu$:

$$\lambda_\eta = Y - \frac{(Y+e^{-\kappa})x}{1+x} = \frac{Y-\mu}{1+x}, \qquad \lambda_{\eta\eta} = -\frac{\mu + Yx}{(1+x)^2}, \qquad \lambda_{\eta\kappa} = -\frac{(Y-\mu)x}{(1+x)^2}.$$

Since λ is a sum of per-voxel terms and S_{pv} enters as η_p does, Σ_p is diagonal with entries $-\lambda_{\eta\eta} = w_{pv}$, and $(\tau_p)_{vq}$ is $-\lambda_{\eta\eta}\partial_q\eta_p - \lambda_{\eta\kappa}\partial_q\kappa_p$. In κ,

$$\lambda_\kappa = Y - e^{-\kappa}\big(\psi(Y+e^{-\kappa}) - \psi(e^{-\kappa})\big) + \mu\,g(x) - \frac{Yx}{1+x}.$$

For integer Y, $\psi(Y+r) - \psi(r) = \sum_{j<Y} 1/(r+j)$, so the first two terms are $\sum_{j<Y} jk/(1+jk)$; summed over voxels this is G_{1p}. Differentiating once more, and using $\psi_1(Y+r) - \psi_1(r) = -\sum_{j<Y} 1/(r+j)^2$, gives $\Lambda_{\kappa\kappa}$ with G_{2p}. Since γ enters f_p only through (η_p, κ_p), the chain rule gives (17), the last two terms being the curvature of the inner map. □

Lemma 7 (Derivatives of η_p and κ_p)**.** *Let $U_q = \sum_{i\in\mathcal{I}_p} e_i G_{iq}$, $W_q = \sum_{i\in\mathcal{I}_p} e_i^2 G_{iq}$, $U^{(2)}_{qq'} = \sum_{i\in\mathcal{I}_p} e_i G_{iq}G_{iq'}$, $W^{(2)}_{qq'} = \sum_{i\in\mathcal{I}_p} e_i^2 G_{iq}G_{iq'}$, and*

$$\ell_q = \frac{U_q}{s_1}, \quad h_q = \frac{2W_q}{s_2}, \quad \ell^{(2)}_{qq'} = \frac{U^{(2)}_{qq'}}{s_1} - \frac{U_qU_{q'}}{s_1^2}, \quad h^{(2)}_{qq'} = \frac{4W^{(2)}_{qq'}}{s_2} - \frac{4W_qW_{q'}}{s_2^2}.$$

Then

$$\partial_q\eta_p = \ell_q, \qquad \partial_q\kappa_p = h_q - 2\ell_q, \qquad \partial^2_{qq'}\eta_p = \ell^{(2)}_{qq'}, \qquad \partial^2_{qq'}\kappa_p = h^{(2)}_{qq'} - 2\ell^{(2)}_{qq'}.$$

Proof. $\eta_p = \log s_1$ and $\kappa_p = \log\theta_p + \log s_2 - 2\log s_1$. Since $\partial_q s_1 = U_q$ and $\partial_q s_2 = 2W_q$, we get $\partial_q \log s_1 = \ell_q$ and $\partial_q \log s_2 = h_q$; differentiating again, $\partial^2_{qq'}\log s_1 = \ell^{(2)}_{qq'}$ and $\partial^2_{qq'}\log s_2 = h^{(2)}_{qq'}$ (the factor 4 arises because $\partial^2_{qq'}s_2 = 4W^{(2)}_{qq'}$). □

Remark 8 (Why (η,κ))**.** An earlier form of this derivation worked in $R_p = s_1^2/(\theta_p s_2) = 1/k_p$ and $A_p = s_1/(\theta_p s_2)$. It is the same mathematics, but not the same arithmetic. As $\theta_p \to 0$ both grow like $1/\theta_p$, and the moderator block subtracts terms of that order that should cancel to order one. On a 1,328-experiment voxel-based meta-regression, one pattern of 99 experiments fitted $\theta_p = 8.4\times10^{-13}$, so $r \approx 1.6\times10^{12}$. For that pattern's diagonal moderator entry a 50-digit evaluation gives 30.92; the (R,A) form gave 5.89, and the information had a negative eigenvalue. In (η,κ) every term of Proposition 6 is of order one: w, c and the summands of G_{1p}, G_{2p} are bounded, and $g(x) \approx x/2$ is summed from its power series $\sum_{n\geq1}(-1)^{n+1}nx^n/(n+1)$ below $x = 10^{-2}$, where the direct form cancels. The (η,κ) form agrees with the 50-digit value to 10^{-7} relative. Autodiff gave 27.69: it differentiates the likelihood as NiMARE writes it, $\log\Gamma(Y+r) - \log\Gamma(r)$ with $r \approx 10^{12}$, which is not exact there either.

The sums (16) need integer counts. CBMR's foci are integers, so their pattern marginals are too; for fractional counts the implementation uses the digamma forms, which are exact away from $k_p = 0$.

Corollary 9 (Poisson as a limit)**.** *As $\theta_p \to 0$ with γ fixed, $x_{pv} \to 0$ and*

$$w_{pv} = \frac{\mu_{pv} + Y_{pv}x_{pv}}{(1+x_{pv})^2} \;\longrightarrow\; s_1 u_{pv} \;=\; T_p u_{pv},$$

recovering Σ_p of Proposition 3. In particular the negative binomial weight is foci-dependent at any $\theta_p > 0$ and the dependence vanishes only in the Poisson limit. In the same limit c_{pv}, G_{1p}, G_{2p} and $\mu_{pv}g(x_{pv})$ vanish, so (17) *tends to the information of a Poisson likelihood on the pattern marginals. That differs from the $\gamma\gamma$ block of Proposition 3, whose count term $\sum_i y_{i\cdot}m_i$ is taken per experiment, by $-\big(\sum_v Y_{pv}\big)\ell^{(2)}_{qq'}$.*

5 Clustered Negative Binomial

5.1 The likelihood

Here the latent effect is a property of the experiment rather than of the voxel, so the spatial coefficients enter only through the *scalar* $E_p = \sum_v u_{pv}$. With precision $\nu_p = 1/\theta_p$ and $n_p = |\mathcal{I}_p|$,

$$\begin{aligned}\log\mathcal{L}_p = {} & n_p\nu_p\log\nu_p - n_p\log\Gamma(\nu_p) + \sum_{i\in\mathcal{I}_p}\log\Gamma(y_{i\cdot}+\nu_p) \\ & - \sum_{i\in\mathcal{I}_p}(y_{i\cdot}+\nu_p)\log\big(E_p e_i + \nu_p\big) + \sum_v Y_{pv}S_{pv} + \sum_{i\in\mathcal{I}_p} y_{i\cdot}m_i.\end{aligned} \tag{19}$$

Proposition 10 (Clustered negative binomial observed information)**.** *Let $D_i = E_{\pi(i)}e_i + \nu_{\pi(i)}$, $a_{pk} = \sum_v u_{pv}B_{vk}$ and*

$$g'_p = \sum_{i\in\mathcal{I}_p}\frac{(y_{i\cdot}+\nu_p)e_i}{D_i}, \qquad g''_p = -\sum_{i\in\mathcal{I}_p}\frac{(y_{i\cdot}+\nu_p)e_i^2}{D_i^2}.$$

Then Σ_p is diagonal plus rank one,

$$\Sigma_p \;=\; g'_p\,\mathrm{diag}(u_{p\cdot}) \;+\; g''_p\,u_{p\cdot}u_{p\cdot}^\top, \tag{20}$$

and consequently

$$\frac{\partial^2 f_p}{\partial\beta_{ck}\partial\beta_{c'k'}} = L_{pc}L_{pc'}\Big[g'_p\big(B^\top \operatorname{diag}(u_{p\cdot})B\big)_{kk'} + g''_p\, a_{pk}a_{pk'}\Big], \tag{21}$$

$$\frac{\partial^2 f_p}{\partial\beta_{ck}\partial\gamma_q} = L_{pc}\, a_{pk} \sum_{i\in\mathcal{I}_p} \frac{(y_{i\cdot}+\nu_p)\nu_p e_i G_{iq}}{D_i^2}, \tag{22}$$

$$\frac{\partial^2 f_p}{\partial\gamma_q\partial\gamma_{q'}} = \sum_{i\in\mathcal{I}_p} \frac{(y_{i\cdot}+\nu_p)\nu_p E_p e_i}{D_i^2}\, G_{iq}G_{iq'}. \tag{23}$$

Only the experiment totals $y_{i\cdot}$ appear; the voxelwise marginals Y_{pv} enter (19) *linearly and so vanish.*

Proof. Let $\Gamma_p(E,\gamma) = \sum_{i\in\mathcal{I}_p}(y_{i\cdot}+\nu_p)\log(Ee_i+\nu_p)$, so that $f_p = \Gamma_p(E_p,\gamma)$ up to terms linear in ϕ and terms free of ϕ. Since $\partial E_p/\partial S_{pv} = u_{pv}$ and $\partial^2 E_p/\partial S_{pv}\partial S_{pv'} = u_{pv}\delta_{vv'}$,

$$\frac{\partial^2 f_p}{\partial S_{pv}\partial S_{pv'}} = \Gamma''_p(E_p)\, u_{pv}u_{pv'} + \Gamma'_p(E_p)\, u_{pv}\delta_{vv'},$$

and $\Gamma'_p = g'_p$, $\Gamma''_p = g''_p$ by direct differentiation, giving (20). Applying Lemma 1 and using $B^\top(uu^\top)B = (B^\top u)(B^\top u)^\top = a_p a_p^\top$ gives the $\beta\beta$ block. For the cross block, $\partial_q\big[e_i/D_i\big] = \nu_p e_i G_{iq}/D_i^2$, whence $\partial_q\Gamma'_p = \sum_i (y_{i\cdot}+\nu_p)\nu_p e_i G_{iq}/D_i^2$ and the β-derivative contributes the factor $L_{pc}a_{pk}$. For the $\gamma\gamma$ block, $\partial_q\Gamma_p = \sum_i (y_{i\cdot}+\nu_p)E_p e_i G_{iq}/D_i$ and $\partial_{q'}\big[E_p e_i/D_i\big] = E_p e_i \nu_p G_{iq'}/D_i^2$. □

Remark 11. Proposition 10 is the only one of the three whose Σ_p is not diagonal, and it is nonetheless the cheapest to prove: with ν_p fixed, every special function in (19) depends on constants alone.

6 Equivalence

Two independent checks establish that the propositions describe the same matrix NiMARE computes.

6.1 Symbolic

`proofs/negative_binomial.py` and `proofs/clustered_negative_binomial.py` construct the likelihood exactly as `nimare/meta/cbmr/distributions.py` writes it, differentiate it with `sympy`, subtract the claimed closed form, and simplify. Every residual is the symbol 0; nothing numerical is involved, so these are proofs on a symbolic instance rather than evidence at a sample point.

For the clustered case the whole composite Hessian closes in one pass. For the negative binomial the composite is intractable for `simplify`, so the proof is decomposed exactly as the argument above is: (i) NiMARE's summand equals λ of Lemma 5; (ii) the partials of λ are those of Proposition 6, and S enters as η does; (iii) the derivatives of η, κ are those of Lemma 7. Composition is then Lemma 1 together with the chain rule, which is a theorem. Two identities the implementation relies on for precision are checked as well: at integer counts the digamma terms equal the sums (16), and g has the power series of Remark 8.

6.2 Numerical

`scripts/test_cbmr_analytic.py` and `scripts/test_cbmr_designs.py` compare the implementations — vectorisation, pattern assembly, loadings scatter — against

`CBMRModel.information_matrix`, i.e. against `torch.func.hessian` itself. Table 2 reports the design coverage. Two further tests guard the structural claims: the Poisson Hessian must be unchanged when the foci are doubled (Proposition 3), and the other two must move, so a foci-independent implementation of either cannot pass silently. Near $\theta_p = 0$ autodiff is itself inexact (Remark 8), so NiMARE's test of the negative binomial there compares against a 50-digit `mpmath` Hessian of the likelihood instead.

7 Structure of the information matrix

7.1 Sparsity from the design

Proposition 12 (Column separation)**.** *Let* $\mathrm{supp}(p) = \{c : L_{pc} \neq 0\}$ *and let* $\sim$ *be the transitive closure of the relation* $c \sim c' \iff \exists p :\ c, c' \in \mathrm{supp}(p)$*. If* $Q = 0$ *and* $c \not\sim c'$ *then* $H_{(ck),(c'k')} = 0$ *for all* k, k'*. Consequently* H *is block diagonal over the connected components of* $\sim$*, each block of size* $|\cdot| \cdot K$*.*

Proof. Immediate from (5): each summand carries the factor $L_{pc}L_{pc'}$, which vanishes unless both c and c' lie in $\mathrm{supp}(p)$; if they do, then $c \sim c'$. □

Proposition 12 holds for *any* Σ_p and hence for all three distributions — and indeed for any likelihood of the form (4). It is a statement about the design, not about the observation model. Note the entries are *structurally* zero, not small: no rounding decision is involved.

7.2 Which designs separate

The hypothesis $Q = 0$ is not cosmetic, and neither is the connectivity condition. A cell-means spatial factor `s(a)` assigns each pattern exactly one column, so $\sim$ is the identity and there are C blocks; so does a full interaction `s(a:b)`. Every other construction the CBMR grammar admits couples the columns:

- a sum-to-zero term `sz(a)` spans all levels in every row, so $\mathrm{supp}(p)$ is the whole term and one component absorbs everything;
- a spatial covariate `s(x)` loads alongside whatever factor accompanies it;
- any global moderator makes $Q > 0$, and the cross block (6) is generally dense, coupling every spatial column through γ.

Table 2 makes this concrete. The practical consequence is that the closed-form Hessian is general while block-diagonal inversion is not, and the two must be gated separately.

7.3 Bordered structure

When the spatial columns separate but $Q > 0$, H is not block diagonal but is *bordered* block diagonal,

$$H = \begin{pmatrix} D & C \\ C^\top & E \end{pmatrix}, \qquad D = \mathrm{blkdiag}(D_1, \ldots, D_M) \in \mathbb{R}^{CK \times CK}, \quad C \in \mathbb{R}^{CK \times Q}, \tag{24}$$

with Q typically between 1 and 5.

Proposition 13 (Bordered inverse)**.** *If* D *and the Schur complement* $\Sigma = E - C^\top D^{-1} C$ *are nonsingular, then*

$$H^{-1} = \begin{pmatrix} D^{-1} + D^{-1} C\, \Sigma^{-1} C^\top D^{-1} & -D^{-1} C\, \Sigma^{-1} \\ -\Sigma^{-1} C^\top D^{-1} & \Sigma^{-1} \end{pmatrix}, \tag{25}$$

computable in $O\big(\sum_j |D_j|^3\big) + O(CKQ \cdot \max_j |D_j|) + O(Q^3)$ *plus the* $O((CK)^2 Q)$ *cost of forming the outer product, rather than* $O((CK+Q)^3)$.

Proof. Standard block inversion; verify by multiplying out $HH^{-1} = I$ blockwise, using $E - C^\top D^{-1} C = \Sigma$. □

Proposition 13 extends the fast *inverse* to designs of the form "cell-means factor plus global moderators". It does not extend the fast *condition number*: the spectrum of (24) is not the union of the spectra of D and Σ, and no interlacing bound recovers the value.

8 Condition number and inversion

`CBMRModel.covariance` computes $\kappa_2(H) = \sigma_{\max}/\sigma_{\min}$ with `np.linalg.cond`, which forms a full singular value decomposition, and then inverts with `np.linalg.inv`. At the 21-group shape these cost 163 s and 14 s respectively, so the diagnostic dominates the quantity it is diagnosing. Three generalisations are available, in increasing order of applicability.

Proposition 14 (Block-diagonal spectrum)**.** *If* $H = \mathrm{blkdiag}(H_1, \ldots, H_M)$ *then* $\mathrm{spec}(H) = \bigsqcup_j \mathrm{spec}(H_j)$ *and the singular values are likewise the multiset union. Hence* $\kappa_2(H) = \max_j \sigma_{\max}(H_j) \big/ \min_j \sigma_{\min}(H_j)$*, exactly.*

Proof. $H - \lambda I$ is block diagonal with blocks $H_j - \lambda I$, and the determinant of a block diagonal matrix is the product of the blocks' determinants, so the characteristic polynomial factorises. The same argument applied to $H^\top H$ gives the singular values. □

Proposition 15 (Symmetry suffices)**.** *H is symmetric by construction. For symmetric H with eigenvalues λ_i, the singular values are $|\lambda_i|$, so $\kappa_2(H) = \max_i |\lambda_i| / \min_i |\lambda_i|$ and a symmetric eigensolver may replace the SVD. This requires no hypothesis on the design and no positive definiteness.*

Proof. Write $H = V \Lambda V^\top$ with V orthogonal. Then $H^\top H = V \Lambda^2 V^\top$, so the singular values — the nonnegative square roots of the eigenvalues of $H^\top H$ — are $|\lambda_i|$. □

Proposition 16 (One factorisation for both)**.** *If $H \succ 0$, a single Cholesky factorisation $H = LL^\top$ serves both purposes: LAPACK's* `pocon` *returns an estimate of $\kappa_1(H)$ in $O(n^2)$ from L, and* `potri` *returns H^{-1} in $O(n^3/3)$ from the same L. Moreover κ_1 and κ_2 are equivalent within a factor of n:*

$$n^{-1} \kappa_1(H) \;\le\; \kappa_2(H) \;\le\; n\, \kappa_1(H). \tag{26}$$

Proof. The factorisation and the two LAPACK routines are standard. For (26), from $n^{-1/2}\|M\|_1 \le \|M\|_2 \le n^{1/2}\|M\|_1$ applied to $M = H$ and $M = H^{-1}$, $\kappa_2 = \|H\|_2 \|H^{-1}\|_2 \le n^{1/2}\|H\|_1 \cdot n^{1/2}\|H^{-1}\|_1 = n\kappa_1$, and symmetrically. □

Remark 17 (The bound is not the behaviour)**.** Proposition 16 is the fastest route by a wide margin, and (26) makes it sound harmless: a factor of n is four of the sixteen decimal digits. Measured, it is not harmless. On the three structures of Table 3 the `pocon` estimate reports 90.09 against a true κ_2 of 5.02, 2.42×10^5 against 1657.7, and 1312.4 against 4.98 — discrepancies of 18×, 146× and 264×. These are comfortably inside the bound, which is the point: the bound permits behaviour bad enough to make the estimate useless as a drop-in for a number the user sees. A fit at $\kappa_2 = 10^{14}$ — poorly conditioned but still invertible — would be reported at $\sim 10^{16}$ and trip a warning saying its standard errors cannot be trusted.

The conclusion is to split the two uses of the factorisation. The Cholesky is adopted for the *inverse*, where it agrees with `np.linalg.inv` to 1.8×10^{-15} and is 2–3× faster. It is rejected for the *condition number*, which comes instead from Proposition 15: exact, fully general, and still 3–4× faster than the SVD the stock path forms.

8.1 The ladder

Collecting Propositions 13–16 into a dispatch, most specific rung first:

1. **blockwise** — the spatial columns separate and $Q = 0$. Condition number by Proposition 14, inverse blockwise. Reached only by a lone cell-means factor or a lone full interaction.

2. **bordered** — the spatial columns separate but $Q > 0$. Inverse by Proposition 13; the condition number cannot follow, so it falls back to Proposition 15 on the whole matrix.

3. **Cholesky** — no exploitable structure, but $H \succ 0$. Condition number by Proposition 15, inverse by Cholesky.

4. **dense** — $H \not\succ 0$, so the Cholesky declines. Falls through to `np.linalg.inv`, which either succeeds or raises with the diagnostic the stock path would have raised.

Every rung reports the *same* condition number, to the last printed digit; none of them is an estimate. What varies is only how much work is done to obtain it. Structural claims are checked against the matrix in hand rather than assumed — a design that ought to separate but whose information matrix has an entry outside the implied blocks is inverted densely, with a warning, rather than silently truncated.

Table 2 records which rung each design reaches. The fourth is reached by exactly one design in the suite, `~ sz(a) + sz(b) + sz(a:b)`, which is rank deficient at that sample size ($\kappa_2 \approx 4 \times 10^{20}$); the Cholesky declining is the guard working.

9 Empirical results

9.1 Cost of the closed forms

Table 1: Whole `covariance()` call at the 21-group diagnosis shape: 7,329 coefficients, 21 patterns, 8,444 voxels, 349 bases. "chunked jacrev" is $\texttt{jacrev}(\texttt{jacrev}(f), \text{chunk} = 192)$ followed by `np.linalg.cond` and `np.linalg.inv`; stock `torch.func.hessian` does not run at this size at all, requiring a 9.7 GB intermediate. Memory is the increment over the pre-call peak. The last column is the largest relative disagreement between the two covariance matrices.

	chunked jacrev		closed form			
distribution	time	memory	time	memory	speed-up	disagreement
Poisson	205.4 s	+2.29 GB	1.5 s	+0.13 GB	139×	5.3×10^{-15}
NegativeBinomial	343.8 s	+2.74 GB	1.5 s	+0.00 GB	226×	6.3×10^{-15}
ClusteredNegativeBinomial	287.3 s	—	2.3 s	—	124×	3.7×10^{-15}

9.2 Design coverage

9.3 Condition number and inverse

10 Scope and limitations

1. The derivations hold with the overdispersion parameters θ_p held fixed. This matches what `information_matrix` does and what CBMR reports, but it means the reported standard errors do not propagate uncertainty in θ. Extending to the joint (ϕ, θ) information would require the θ blocks, which are not derived here.

Table 2: Design coverage. "patterns" is P and "$\max|\operatorname{supp}(p)|$" the largest number of spatial columns any one pattern loads on — greater than one exactly when the loadings are not one-hot. "blocks" is the number of components of Proposition 12; "rung" is which strategy of Section 8.1 that buys. "disagreement" is the largest relative difference between the closed-form Hessian and `torch.func.hessian`, over the distributions that accept the design. Every row was checked for all three distributions except where noted.

design	formula	P	$\max\|\operatorname{supp}(p)\|$	blocks	rung	disagreement
cell-means factor	`~ s(a)`	4	1	4	blockwise	2.7×10^{-15}
full interaction	`~ s(a:b)`	8	1	8	blockwise	2.7×10^{-15}
additive main effects	`~ sz(a) + sz(b)`	8	5	1	Cholesky	5.0×10^{-16}
main effects + interaction	`~ sz(a) + sz(b) + sz(a:b)`	8	12	1	dense[†]	4.6×10^{-16}
one global moderator	`~ s(a) + x`	4	1	1	bordered	2.2×10^{-14}
global moderator interaction	`~ s(a) + x * z`	4	1	1	bordered	9.2×10^{-14}
voxelwise moderator, continuous	`~ s(a) + s(x)`	96	2	1	Cholesky	2.3×10^{-16}[‡]
voxelwise moderator, coarse	`~ s(a) + s(w)`	12	2	1	Cholesky	4.1×10^{-16}
everything at once	`~ sz(a) + sz(b) + s(w) + x * z`	24	6	1	Cholesky	1.2×10^{-14}

[†] Rank deficient at this sample size (condition number 4×10^{20}), so the Cholesky declines and the ladder falls through to the general inverse. The guard working, not a gap.
[‡] Poisson only: NiMARE's `check_design` refuses the overdispersed distributions when a continuous spatial covariate leaves every pattern with one experiment. The same term with a coarse covariate (previous row) is accepted by all three.

Table 3: Condition number and inverse, on synthetic matrices at the 21-group shape with the three structures real designs produce. "dense" is the stock path (`np.linalg.cond` + `np.linalg.inv`). All inverses agree with the dense one to $\leq 4.6 \times 10^{-15}$ relative. Rows marked $\star$ are what the ladder adopts. The blockwise row was measured with a per-block SVD; the implementation uses a per-block symmetric eigensolver, which returns the same values by Propositions 14 and 15.

structure	strategy	κ reported	κ time	inverse time	total
block diagonal `~ s(a)`, 7329^2	dense (SVD + LU)	5.0243	157.2 s	14.2 s	171.4 s
	`eigvalsh` + Cholesky	5.0243	36.5 s	7.6 s	44.1 s
	Cholesky + `pocon`	90.093	2.9 s	5.5 s	8.4 s
	$\star$ blockwise	5.0243	0.4 s	0.2 s	**0.7** s
bordered `~ s(a) + x * z`, 7331^2	dense (SVD + LU)	1657.7	169.0 s	20.7 s	189.6 s
	$\star$ `eigvalsh` + Schur	1657.7	48.5 s	4.1 s	**52.7** s
	Cholesky + `pocon`	2.42×10^5	4.7 s	8.7 s	13.3 s
	blockwise	*not applicable — the matrix does not separate*			
fully coupled `~ sz(a) + sz(b)`, 7329^2	dense (SVD + LU)	4.9796	161.8 s	19.8 s	181.7 s
	$\star$ `eigvalsh` + Cholesky	4.9796	52.1 s	15.8 s	**67.8** s
	Cholesky + `pocon`	1312.4	4.4 s	8.6 s	13.0 s

2. NiMARE's `check_design` refuses the overdispersed distributions when any pattern has a single experiment, which excludes a continuous spatial covariate. That is a pre-existing restriction of the moment-matching approximation, not of these derivations; the same term with a coarse covariate is accepted and handled.

3. The negative binomial closed form stays accurate as $\theta_p \to 0$ (Remark 8), but the likelihood NiMARE fits is still written in $r = 1/k_p$. A fit that drives θ_p toward zero evaluates $\log\Gamma(Y + r) - \log\Gamma(r)$ with r of order 10^{12}, which loses precision in the fit and in the autodiff Hessian, though not in the closed form.

4. The covariance is returned as a dense $(CK + Q)^2$ array because `contrasts.py` indexes it densely. At the 21-group shape that is 430 MB, and it is the remaining memory floor; exploiting the block structure past the covariance itself would require changes inside NiMARE.

11 Generated proof fragments

Everything below is written by `run_proofs.py` from the same sympy objects the proofs check. Nothing here is transcribed by hand, so the document cannot disagree with the verification that backs it.

11.1 Poisson

The log-likelihood on marginals, dropping the parameter-free $-\sum \log y!$, is $\log \mathcal{L} = \sum_v Y_v S_v + \sum_i y_i m_i - (\sum_v e^{S_v})T$ with $T = \sum_i e^{m_i}$. The first two terms are linear in the coefficients, so they differentiate away entirely; that is the whole of the data dependence.
The expressions the claims are formed from:

$$-\log \mathcal{L} = -Y_0\,(B_{00}\beta_0 + B_{01}\beta_1) - Y_1\,(B_{10}\beta_0 + B_{11}\beta_1) - Y_2\,(B_{20}\beta_0 + B_{21}\beta_1) - y_0\,(G_{00}\gamma_0 + G_{01}\gamma_1) - y_1\,(G_{10}\gamma_0 + G_1$$

Residuals, each asserted to vanish:

claim	residual	result
d2/dbeta0 dbeta0	$\partial^2_{\beta_0\beta_0} - T\,(B^\top \mathrm{diag}(u) B)_{00}$	0
d2/dbeta0 dbeta1	$\partial^2_{\beta_0\beta_1} - T\,(B^\top \mathrm{diag}(u) B)_{01}$	0
d2/dbeta1 dbeta0	$\partial^2_{\beta_1\beta_0} - T\,(B^\top \mathrm{diag}(u) B)_{10}$	0
d2/dbeta1 dbeta1	$\partial^2_{\beta_1\beta_1} - T\,(B^\top \mathrm{diag}(u) B)_{11}$	0
d2/dbeta0 dgamma0	$\partial^2_{\beta_0\gamma_0} - a_0 U_0$	0
d2/dbeta0 dgamma1	$\partial^2_{\beta_0\gamma_1} - a_0 U_1$	0
d2/dbeta1 dgamma0	$\partial^2_{\beta_1\gamma_0} - a_1 U_0$	0
d2/dbeta1 dgamma1	$\partial^2_{\beta_1\gamma_1} - a_1 U_1$	0
d2/dgamma0 dgamma0	$\partial^2_{\gamma_0\gamma_0} - E\sum_i e_i G_{i0} G_{i0}$	0
d2/dgamma0 dgamma1	$\partial^2_{\gamma_0\gamma_1} - E\sum_i e_i G_{i0} G_{i1}$	0
d2/dgamma1 dgamma0	$\partial^2_{\gamma_1\gamma_0} - E\sum_i e_i G_{i1} G_{i0}$	0
d2/dgamma1 dgamma1	$\partial^2_{\gamma_1\gamma_1} - E\sum_i e_i G_{i1} G_{i1}$	0
Hessian contains no foci symbol	$\{Y_v, y_i\} \cap \mathrm{free}(H)$	0

Verdict: every residual is exactly zero.

11.2 Negative Binomial

NiMARE's negative binomial uses shape $r = s_1^2/(\theta s_2)$ and success probability $\pi_v = (1 + s_1/(\theta u_v s_2))^{-1}$. Writing $\eta = \log s_1$ and $\kappa = \log k$ with $k = \theta s_2/s_1^2$, the voxel mean is $\mu_v = e^\eta u_v$, the size is $1/k$, and each voxel's summand depends on (η, κ) and S_v alone. Below, $x_v = k\mu_v$.
The expressions the claims are formed from:

$$\lambda(\eta, \kappa, S_v) = \kappa y + y \log\left(e^S e^\eta\right) - \left(y + e^{-\kappa}\right) \log\left(e^S e^\eta e^\kappa + 1\right) + \mathrm{loggamma}\left(y + e^{-\kappa}\right) - \mathrm{loggamma}\left(e^{-\kappa}\right)$$

One voxel's log-likelihood, up to $-\log\Gamma(Y_v + 1)$.

$$G_1 = y - \left(\mathrm{polygamma}\left(0, y + e^{-\kappa}\right) - \mathrm{polygamma}\left(0, e^{-\kappa}\right)\right) e^{-\kappa}$$

The count term of λ_κ. polygamma$(0, \cdot)$ is the digamma function.

$$G_2 = \left(\mathrm{polygamma}\left(0, y + e^{-\kappa}\right) - \mathrm{polygamma}\left(0, e^{-\kappa}\right)\right) e^{-\kappa} + \left(\mathrm{polygamma}\left(1, y + e^{-\kappa}\right) - \mathrm{polygamma}\left(1, e^{-\kappa}\right)\right) e$$

The count term of $\lambda_{\kappa\kappa}$.

$$g(x_v) = -\frac{1}{x_v+1} + \frac{\log(x_v+1)}{x_v}$$

Residuals, each asserted to vanish:

claim	residual	result
lambda(eta,kappa,S) equals NiMARE's summand	$\lambda\big\|_{\eta,\kappa} - \left(\log \mathcal{L}_v + \log\Gamma(Y_v+1)\right)$	0
d lambda/dS = d lambda/deta	$\partial_{S_v}\lambda - \partial_\eta\lambda$	0
lambda_eta	$\lambda_\eta - \frac{Y_v-\mu_v}{1+x_v}$	0
lambda_eta_eta	$\lambda_{\eta\eta} + \frac{\mu_v+Y_v x_v}{(1+x_v)^2}$	0
lambda_eta_kappa	$\lambda_{\eta\kappa} + \frac{(Y_v-\mu_v)x_v}{(1+x_v)^2}$	0
lambda_kappa	λ_κ as implemented	0
lambda_kappa_kappa	$\lambda_{\kappa\kappa}$ as implemented	0
count sum in lambda_kappa, y=0	$G_1(Y_v=0) - \sum_{j<0}\frac{jk}{1+jk}$	0
count sum in lambda_kappa_kappa, y=0	$G_2(Y_v=0) - \sum_{j<0}\frac{jk}{(1+jk)^2}$	0
count sum in lambda_kappa, y=1	$G_1(Y_v=1) - \sum_{j<1}\frac{jk}{1+jk}$	0
count sum in lambda_kappa_kappa, y=1	$G_2(Y_v=1) - \sum_{j<1}\frac{jk}{(1+jk)^2}$	0
count sum in lambda_kappa, y=2	$G_1(Y_v=2) - \sum_{j<2}\frac{jk}{1+jk}$	0
count sum in lambda_kappa_kappa, y=2	$G_2(Y_v=2) - \sum_{j<2}\frac{jk}{(1+jk)^2}$	0
count sum in lambda_kappa, y=3	$G_1(Y_v=3) - \sum_{j<3}\frac{jk}{1+jk}$	0
count sum in lambda_kappa_kappa, y=3	$G_2(Y_v=3) - \sum_{j<3}\frac{jk}{(1+jk)^2}$	0
count sum in lambda_kappa, y=4	$G_1(Y_v=4) - \sum_{j<4}\frac{jk}{1+jk}$	0
count sum in lambda_kappa_kappa, y=4	$G_2(Y_v=4) - \sum_{j<4}\frac{jk}{(1+jk)^2}$	0
count sum in lambda_kappa, y=5	$G_1(Y_v=5) - \sum_{j<5}\frac{jk}{1+jk}$	0
count sum in lambda_kappa_kappa, y=5	$G_2(Y_v=5) - \sum_{j<5}\frac{jk}{(1+jk)^2}$	0
gap series to xˆ10	$\left[\frac{\log(1+x)}{x} - \frac{1}{1+x}\right]_{x^10} - \sum_{n=1}^{10}\frac{(-1)^{n+1}nx^n}{n+1}$	0
deta/dgamma0	$\partial_{\gamma_0}\eta - \ell_0$	0
dkappa/dgamma0	$\partial_{\gamma_0}\kappa - (h_0 - 2\ell_0)$	0
d2eta/dgamma0 dgamma0	$\partial^2_{\gamma_0\gamma_0}\eta - \ell^{(2)}_{00}$	0
d2kappa/dgamma0 dgamma0	$\partial^2_{\gamma_0\gamma_0}\kappa - (h^{(2)}_{00} - 2\ell^{(2)}_{00})$	0
d2eta/dgamma0 dgamma1	$\partial^2_{\gamma_0\gamma_1}\eta - \ell^{(2)}_{01}$	0
d2kappa/dgamma0 dgamma1	$\partial^2_{\gamma_0\gamma_1}\kappa - (h^{(2)}_{01} - 2\ell^{(2)}_{01})$	0
deta/dgamma1	$\partial_{\gamma_1}\eta - \ell_1$	0
dkappa/dgamma1	$\partial_{\gamma_1}\kappa - (h_1 - 2\ell_1)$	0
d2eta/dgamma1 dgamma0	$\partial^2_{\gamma_1\gamma_0}\eta - \ell^{(2)}_{10}$	0
d2kappa/dgamma1 dgamma0	$\partial^2_{\gamma_1\gamma_0}\kappa - (h^{(2)}_{10} - 2\ell^{(2)}_{10})$	0
d2eta/dgamma1 dgamma1	$\partial^2_{\gamma_1\gamma_1}\eta - \ell^{(2)}_{11}$	0
d2kappa/dgamma1 dgamma1	$\partial^2_{\gamma_1\gamma_1}\kappa - (h^{(2)}_{11} - 2\ell^{(2)}_{11})$	0

Verdict: every residual is exactly zero.

11.3 Clustered Negative Binomial

The latent effect is a property of the experiment rather than of the voxel, so the spatial coefficients reach the likelihood only through the scalar $E = \sum_v u_v$. Writing $\Gamma(E) = \sum_i (y_i+\nu)\log(Ee_i+\nu)$, the second differential in S is $\Gamma''(E)u_v u_{v'} + \Gamma'(E)u_v\delta_{vv'}$ — diagonal plus rank one.
The expressions the claims are formed from:

$$-\log\mathcal{L}_p = -Y_0\left(B_{00}\beta_0 + B_{01}\beta_1\right) - Y_1\left(B_{10}\beta_0 + B_{11}\beta_1\right) - Y_2\left(B_{20}\beta_0 + B_{21}\beta_1\right) - 2\nu\log(\nu) - y_0\left(G_{00}\gamma_0 + G_{01}\gamma_1\right) - y_1$$

$$g' = \frac{(\nu + y_0)\, e^{G_{00}\gamma_0 + G_{01}\gamma_1}}{\nu + \left(e^{B_{00}\beta_0 + B_{01}\beta_1} + e^{B_{10}\beta_0 + B_{11}\beta_1} + e^{B_{20}\beta_0 + B_{21}\beta_1}\right) e^{G_{00}\gamma_0 + G_{01}\gamma_1}} + \frac{(\nu + y_1)\, e^{G_{10}\gamma_0 + G_{11}\gamma_1}}{\nu + (e^{B_{00}\beta_0 + B_{01}\beta_1} + e^{B_{10}\beta_0 + B_{11}\beta_1} + e^{B_{20}\beta_0 +}}$$

$$g'' = -\frac{(\nu + y_0)\, e^{2G_{00}\gamma_0 + 2G_{01}\gamma_1}}{\left(\nu + \left(e^{B_{00}\beta_0 + B_{01}\beta_1} + e^{B_{10}\beta_0 + B_{11}\beta_1} + e^{B_{20}\beta_0 + B_{21}\beta_1}\right) e^{G_{00}\gamma_0 + G_{01}\gamma_1}\right)^2} - \frac{(\nu + y_1)\, e^{2G_{10}\gamma_0 +}}{(\nu + (e^{B_{00}\beta_0 + B_{01}\beta_1} + e^{B_{10}\beta_0 + B_{11}\beta_1} + e}$$

Residuals, each asserted to vanish:

claim	residual	result
d2/dbeta0 dbeta0	$\partial^2_{\beta_0\beta_0}$ − closed form	0
d2/dbeta0 dbeta1	$\partial^2_{\beta_0\beta_1}$ − closed form	0
d2/dbeta0 dgamma0	$\partial^2_{\beta_0\gamma_0}$ − closed form	0
d2/dbeta0 dgamma1	$\partial^2_{\beta_0\gamma_1}$ − closed form	0
d2/dbeta1 dbeta1	$\partial^2_{\beta_1\beta_1}$ − closed form	0
d2/dbeta1 dgamma0	$\partial^2_{\beta_1\gamma_0}$ − closed form	0
d2/dbeta1 dgamma1	$\partial^2_{\beta_1\gamma_1}$ − closed form	0
d2/dgamma0 dgamma0	$\partial^2_{\gamma_0\gamma_0}$ − closed form	0
d2/dgamma0 dgamma1	$\partial^2_{\gamma_0\gamma_1}$ − closed form	0
d2/dgamma1 dgamma1	$\partial^2_{\gamma_1\gamma_1}$ − closed form	0

Verdict: every residual is exactly zero.

11.4 Covariance strategies

Let $\mathrm{supp}(p) = \{c : L_{pc} \neq 0\}$ and let $\sim$ be the transitive closure of $c \sim c' \iff \exists p :\ c, c' \in \mathrm{supp}(p)$. Every distribution's Hessian has $\beta\beta$ block $\sum_p L_{pc} L_{pc'} (B^\top \Sigma_p B)_{kk'}$, so the factor $L_{pc} L_{pc'}$ decides which entries can be nonzero at all.
Residuals, each asserted to vanish:

claim	residual	result
columns sharing no pattern have a zero cross block	$H_{(0k),(2k')}$ with $0 \not\sim 2$	0
characteristic polynomial factorises over the blocks	$\det(\mathrm{blkdiag}(A, B) - \lambda I) - \det(A - \lambda I)\det(B - \lambda I)$	0
symmetric H satisfies H^T H = H^2	$H^\top H - H^2$	0
the singular values are the absolute eigenvalues	$\det(H^2 - \lambda^2 I) - \det(H - \lambda I)\det(H + \lambda I)$	0
blockwise inverse	$\mathrm{blkdiag}(A, B)^{-1} - \mathrm{blkdiag}(A^{-1}, B^{-1})$	0
D.TL + C.BL = I	$D\,\mathrm{TL} + C\,\mathrm{BL} - I_m$	0
D.TR + C.BR = 0	$D\,\mathrm{TR} + C\,\mathrm{BR}$	0
C^T.TL + E.BL = 0	$C^\top \mathrm{TL} + E\,\mathrm{BL}$	0
C^T.TR + E.BR = I	$C^\top \mathrm{TR} + E\,\mathrm{BR} - I_q$	0

Verdict: every residual is exactly zero.